

EE-1005: Digital Logic Design

Serial No:

Sessional II Exam

Total Time: 1 Hours

Total Marks: 42

Tuesday, 11th April, 2023

Course Instructors

Dr. Mehwish Hassan, Dr. Niaz Ahmad

Shehzad Ahmad, Dr. Muhammad Awais, Nirmal

Tariq

Signature of Invigilator

Solution.

Student Name

Roll No.

Section

Signature

DO NOT OPEN THE QUESTION BOOK OR START UNTIL INSTRUCTED.

Instructions:

1. Attempt on question paper. Attempt all of them. No additional sheet will be provided for rough work.
2. After asked to commence the exam, please verify that you have **Three (03) questions printed on seven (07) pages** including this title and rough work pages.
3. Read the questions carefully for clarity of context and understanding of meaning and make assumptions wherever required, for neither the invigilator will address your queries, nor the teacher/examiner will come to the examination hall for any assistance.
4. The exam is closed books, closed notes. Please see that the area in your threshold is free of any material classified as 'useful in the paper' or else there may a charge of cheating.
5. Calculator sharing is strictly prohibited.
6. Use permanent ink pens only. **Any part done using soft pencil will not be marked and cannot be claimed for rechecking.**

	Q-1	Q-2	Q-3	Total
Marks Obtained				
Total Marks	10	16	16	42

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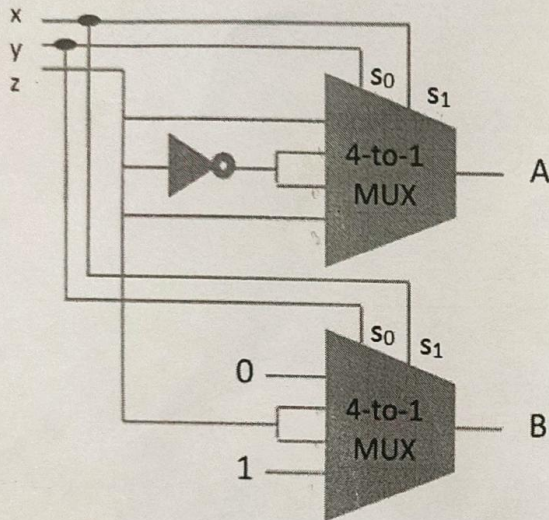
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Question 1 [10 Marks]

I. Determine the outputs A and B as functions of inputs x, y and z. [8]



S ₁ S ₀		Sum Carry		
x	y	z	A	B
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

II. What device is implemented using multiplexers? [2]

It is full adder with 'A' represents sum bit while 'B' represents carry.

Question 2 [16 Marks]

1. A four digit multiplexed 7-segment display has seven (7) input lines and four (4) enables as shown in Fig 2.1. The purpose of each enable line is to enable the corresponding 7-segment display. Only one enable line should remain active at any time to display a BCD digit on a corresponding 7-segment display. Design the combinational circuit (**at block level**) so that when select = '00', the four bit BCD number appearing at I_0 is displayed at segment 1, when select = '01', the four bit BCD number appearing at I_1 is displayed at segment 2 and so on. [10]

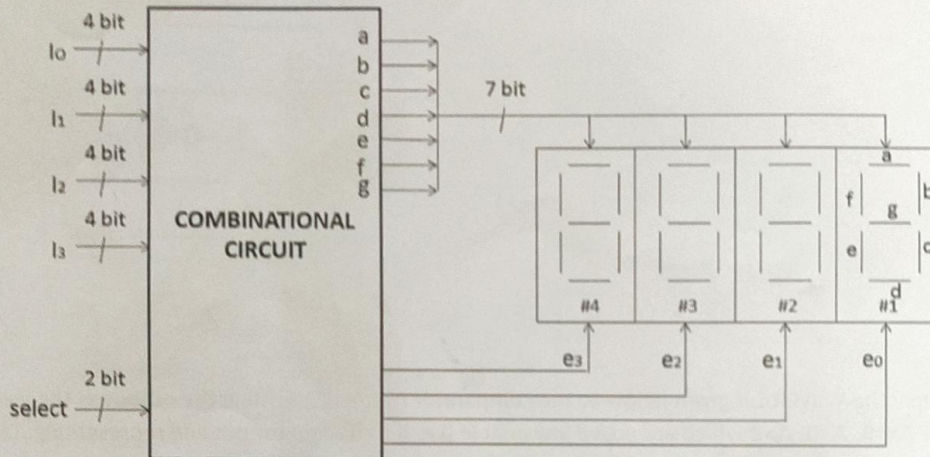
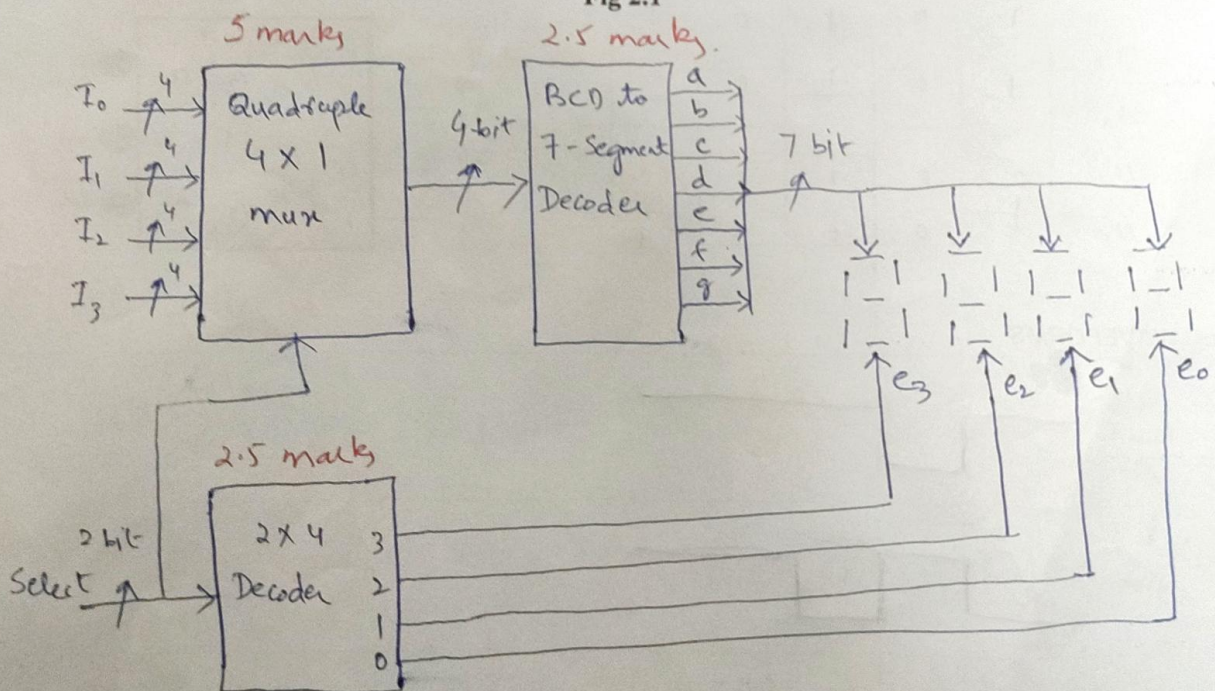
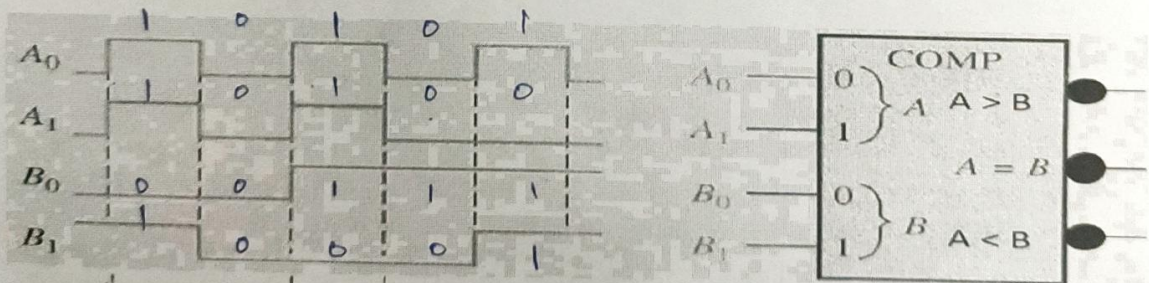


Fig 2.1

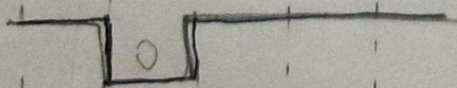


- II. Apply the Waveform given below to the comparator circuit. Determine the output as the waveforms for $A=B$, $A<B$, $A>B$ which are active low enable (i.e. if $A>B$, then output line representing $A>B$ will be low while others will be high). [6]

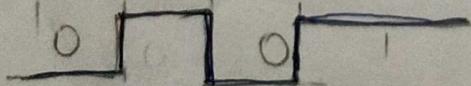


WAVEFORMS

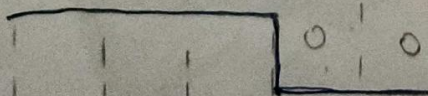
$F_{A=B} =$



$F_{A>B} =$



$F_{A<B} =$



Question 3 [16 Marks]

A small corporation has 100 shares of stock, and each share entitles its owner to one vote at a stockholder's meeting. Ms. Ayesha owns 10 shares, Mr. Basharat owns 20 shares, Mr. Cecil owns 30 shares, and Mr. Danish owns 40 shares. A two-thirds majority is required in order to pass a measure at a stockholder's meeting. Each of the four stockholders has a switch which he or she closes to vote yes for all of his or her shares and opens to vote no. A switching circuit is to be designed to turn on a light if the measure passes.

Note: If the truth table is incorrect, rest of parts (II & III) will not be checked.

- i. Derive a truth table for the output function (Z). (Z = 1 for light on and Z=1 when two-thirds votes are yes i.e. '1') [8]

A(10)	B(20)	C(30)	D(40)	Z	
0	0	0	0	0	
0	0	0	1	0	
0	0	1	0	0	
0	0	1	1	1	✓
0	1	0	0	0	
0	1	0	1	0	
0	1	1	0	0	
0	1	1	1	1	✓
1	0	0	0	0	
1	0	0	1	0	
1	0	1	0	0	
1	0	1	1	1	✓
1	1	0	0	0	
1	1	0	1	1	✓
1	1	1	0	0	
1	1	1	1	1	✓

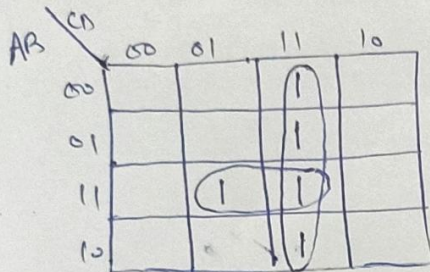
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- II. Write the simplified expression for Z in sum-of-products form using K-map. [4]



$$Z = ABD + CD$$

- III. Draw the logic diagram with NAND gates only using expression developed in part (II). [4]

